# Determinex — Track 2 Project Specification

**2026 AMD Radeon GPU Hackathon · Track 2: Development & Local Deployment of Private AI Agents**

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**Source:** github.com/DarthCeltic/Determinex · **Models:** huggingface.co/darthceltic85 · **Author:** github.com/DarthCeltic

## 1. Application Scenarios

### The problem

Cloud AI coding tools require sending proprietary source to a third party, with no cryptographic guarantee about retention, training use, or jurisdiction. For regulated industries and IP-sensitive work that is not a trade-off, it is a disqualifier.

Local alternatives trade that risk for a worse one: a single small model with no verification layer. A 1.5B model proposing wrong code *confidently* is more dangerous than no assistant, because the reviewer's attention is the scarce resource.

Determinex refuses both. It runs locally on the developer's own hardware — including AMD Radeon GPUs — and it never trusts a model's self-assessment. A real compiler or a real test suite is the only judge.

### Who it is for

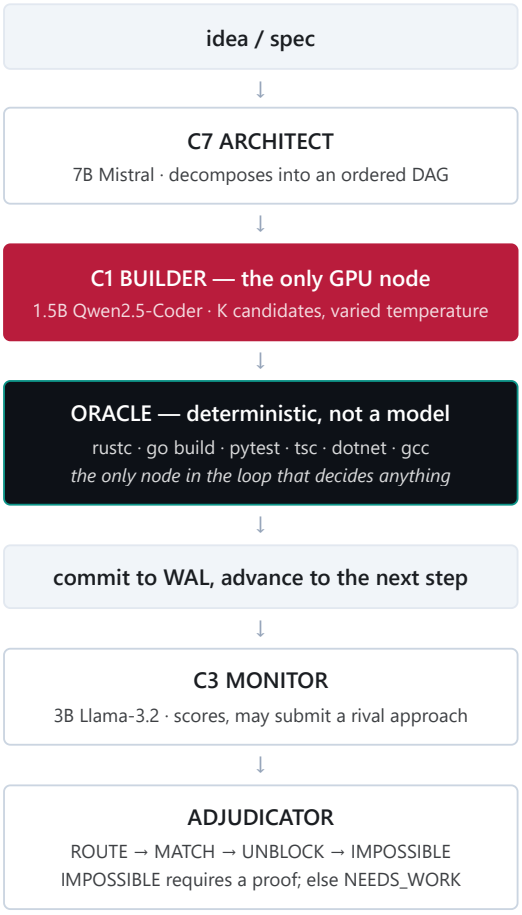
| Scenario | How Determinex handles it |
|---|---|
| Build a program from a plain-language idea | Synthesizes a *sound* test oracle first, then drives verified search until a program passes it |
| Repair a broken repository | Ingests the project, runs its real oracle, assigns honest blame: CODE / ENVIRONMENT / TEST |
| Use a cloud model without exposing source | Project Cloak obfuscates identifiers via AST rewriting before any outbound call |
| Evaluate a local model honestly | Compiler-validated probe harness; no LLM-as-judge anywhere in the loop |

### Why this matters on AMD hardware specifically

Determinex's accuracy is a function of how many candidates it can verify, not of how good any single generation is. That makes it one of the few agent architectures where **GPU batch throughput converts directly into correctness** — quantified in §5.

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## 2. Agent Architecture




**The red node is the only one that touches the GPU, and the black node is the only one that decides anything.** No model ever judges its own output: the Builder proposes K candidates, the Oracle disposes of them with a compiler or a real test suite, and a failure returns the *exact* compiler error to the next attempt rather than a summary of it. That loop is why GPU batch throughput and agent correctness are the same measurement.

### The amplification mechanism

```
per-attempt success probability   p
K candidates sampled, each verified against a sound oracle

    P(at least one correct) = 1 - (1 - p)^K
```

Any `p > 0` is driven toward correct by increasing K. The model supplies search; the oracle supplies truth. **The soundness of the oracle is load-bearing** — a garbage oracle yields confident garbage — which is why every synthesized oracle is validated to actually execute, and why an oracle that can only assert "the symbol exists" is rejected as vacuous rather than reported as a pass.

### The design rule this project is built on

*A check must never report an outcome it did not establish.* It is why `solved` is never claimed without a passing `OracleResult` and a proof, why every guard in this repository is negative-controlled against the defect it exists to catch, and why the boundaries in §6 are stated rather than omitted.

## 3. Core Capabilities

### 3.0 The five capabilities, named

Track 2 asks for **at least two** of five capabilities. Determinex implements **all five**. Each is listed here with the single place in this repository where it can be checked, because a capability that cannot be pointed at is a claim rather than a feature.

| Capability | In Determinex | Where to verify |
|---|---|---|
| **Local knowledge retrieval (RAG)** | A corpus of **234 recorded entries** — including **89 generalized failure classes** and **272 learned solve classes**, each carrying its method, measured numbers and the traps that produce false readings — indexed for hybrid lexical + embedding retrieval over the corpus and the project's own documentation. The agent queries it *before* acting, so accumulated experience is applied first-shot rather than rediscovered. | `python scripts/determinex_corpus_api.py ask "<question>"` ; in the IDE, **Learning Studio → Ask the Corpus** |
| **Tool invocation** | The agent's verdicts come from real tools, never from a model's opinion: `cargo` , `go build` , `tsc --noEmit` , `pytest` , `dotnet build` , `gcc` , plus `git` and Docker. Nineteen language oracles behind one registry. | `scripts/determinex_oracle.py` ; `tests/test_oracle_runtimes_discriminate.py` proves each accepts correct code and rejects broken code |
| **Multi-step task planning** | An Architect decomposes a request into a dependency-ordered DAG of build steps; a Builder executes one step at a time against the current state, and each step is gated by the compiler before the next begins. | `scripts/determinex_hive.py generate-dag` ; `scripts/determinex_decompose.py` |
| **Local multi-turn memory** | Two kinds. Within a session, the Concept Lab interview carries every answer forward and reasons over the accumulated context (§3.0a). Across sessions, Case Memory stores **only oracle-verified** solves, so the system never learns from its own unverified guesses. | `scripts/determinex_case_memory.py` ; session WAL under `sessions/` |
| **Permission control & privacy** | Model-generated code executes only inside `intake.hardened_runner` — workspace-bounded, environment-scrubbed, network and Docker denied by default. Project Cloak obfuscates every identifier before any cloud call. Source mutation is refused unless explicitly authorised. | `scripts/intake/hardened_runner.py` ; `docs/policy/CLOAK_THREAT_MODEL.md` ; `scripts/dev/parallel_execution_layer_audit.py` reports 523 execution sites, 0 unsafe, 0 unclassified |

### 3.0a Multi-turn interaction: the interview ends when the system knows enough, not when a counter runs out

The agent's first job is to find out what you actually want, and this is where most tools quietly fail: they ask a fixed number of questions and then generate.

Determinex asked four questions from a static bank and then wrote the spec — whatever those four answers happened to contain. A second path was worse: it forced readiness after **two** replies, with a comment in the source admitting why ("the 3B model reliably loops on the 3rd exchange; cut it off here"). That trades your specification for the model's stamina, and a spec built from two answers to a five-answer problem produces a confident build of the wrong thing — which the compiler will happily certify, because compiling is not the same as being what you asked for.

Readiness is now **measured**, against this project's own standard: *can a sound oracle be synthesized from what the user has told us?* `determinex_synthesize` answers exactly that and emits an explicit marker when it cannot. Tying the interview to that verdict means the agent's idea of "enough" cannot drift from the verifier's.

Each still-missing requirement maps to a specific question, so follow-ups are derived from what is **absent** rather than drawn from a list — which is also why it cannot loop: a question can only be asked while the thing it asks for is genuinely missing, an already-asked question is never repeated, and a round that establishes nothing new is reported as stalled so the interface offers to build rather than rephrasing forever.

Measured against a realistic user who never volunteers a concrete example until asked:

```
Q1  "It should read a list of bookings and collapse the ones that overlap"   -> keep going
Q2  "Python"                                                                 -> keep going
Q3  "A CLI command I run in the terminal"                                    -> keep going
Q4  "Standard library only, no dependencies"                                 -> keep going
        ^ the old build stopped here and wrote the spec
Q5  "If two bookings just touch end-to-start they should still merge"        -> keep going
Q6  "merge([(9,10),(10,11)]) == [(9,11)]"                                    -> SUFFICIENT
        a sound oracle can be synthesized: 6/6 requirements covered
```

At question four — where it used to give up — the context was still missing the one thing that makes a build verifiable: a worked example. The interface says so on screen while it asks ("Still needed for a verifiable build: example, edge_cases"), so an open-ended interview never reads as an interrogation.

This runs on the Radeon-hosted model, and it is visible end to end in the demo video.

### 3.0b Several AIs in one room, and a foreman who decides who is right

Determinex hosts whichever coding-agent CLIs a developer already has installed, plus a local Ollama model, **in one shared conversation over one workspace**. Turns run one at a time, never concurrently, and **each agent's edits are checked against the real compiler/test oracle before the next agent goes**. All participants read the same Mission Plan, re-synced from the repository's own project documentation, so nobody duplicates or contradicts another's work.

The point is that it does not matter which agents they are. Nothing in the loop knows or cares which one produced a candidate — the oracle is the only thing that decides, so a local 1.5B model and a frontier CLI are judged by exactly the same standard.

Serialising turns prevents a *collision*. It does not answer the two questions that decide whether a room finishes:

> **When two participants disagree, who wins? When nobody is progressing, who goes next?**

Without answers a room degrades the way rooms do: the last agent to speak wins by default — recency wearing the costume of authority — agents politely defer to each other, and the work stops with everyone still "working".

**Authority is what a claim is backed by, never who made it.**

| Tier | Backed by | Why it ranks there |
|---|---|---|
| **ORACLE** | the compiler or tests passed | a fact about the workspace, not an opinion about it |
| **CORPUS** | Determinex's own verified lessons | cannot be outvoted by an agent's confidence |
| **REFUTED** | the oracle rejected it | "this fails with these 3 errors" is *evidence*; a room that discards it re-proposes the same thing forever |
| **PROSE** | an agent talking | lowest, however fluent |

Within a tier the later claim wins, because the later one saw the earlier. **Across tiers time is irrelevant** — a passing oracle result from turn 3 outranks beautiful prose from turn 40 and always will. Status is read from the turn *record*, never from its text: an agent writing "fixed it" is PROSE whatever it claims.

Stall detection reuses the same rule that governs the amplifier's rounds — the room's failing -check count over turns *is* an error series — so the foreman inherits "keep going while the number falls, and say so when the remainder is out of proportion" rather than inventing a second story about when to stop. It then either hands the floor to a participant who has *not* been failing at it, or escalates

naming what is actually needed: a missing toolchain, a stronger model, or a person.

Verify: `python scripts/determinex_agent_chat.py foreman <session-id>` · `scripts/determinex_foreman.py` · 17 tests in `tests/test_foreman_authority.py`

### 3.0c It tells you when it needs you — on the desktop and on your phone

A verified build is long-running by construction: K candidates, an oracle run per candidate, retries. The user is not watching, and that is the normal case rather than the exception.

Determinex raises a **native desktop notification** — Windows, macOS, or Linux, with **no configuration at all** — and additionally posts to Discord, Slack, or Telegram when a webhook is set. Both, always: you may be at the machine in another window, or away from it entirely.

The restraint matters more than the plumbing. Only two situations earn an interruption:

- **DONE** — the work reached a verdict, verified or not. It is over and you can decide.
- **BLOCKED** — it needs a person, which the foreman identifies precisely rather than by timeout.

Everything else is a status you can pull. A stream of normal-progress pings is exactly how the one that mattered gets swiped away.

Verify: `python scripts/determinex_cockpit.py status` · `python scripts/determinex_cockpit.py notify --if-needed`

**3.1 Compiler-verified build loop.** `rustc`, `go build`, Python (syntax → import → test discovery), and `tsc --noEmit` in a purpose-built container. Languages without a configured oracle **fail closed** with an actionable message — they are never silently passed. Every oracle refuses an empty workspace, because a build step whose patch went nowhere would otherwise be recorded as verified.

**3.2 Multi-agent hive.** Three specialists exchange a structured DSL rather than prose.

**3.3 Project Cloak.** AST-aware identifier obfuscation across 10 languages, so a cloud model can assist without seeing proprietary names. Stated precisely: audits confirm that *everything Cloak obfuscated stayed obfuscated*. Extraction completeness is a separate property with its own test — which has caught real gaps, including JavaScript instance fields written `this.field = 0` surviving obfuscation until they were fixed. The distinction is documented rather than smoothed over.

### 3.3a Repairing a real repository — end to end, on the Radeon GPU

The clearest demonstration of application value is not a synthetic task. Determinex was pointed at `mwaskom__seaborn-3010` from SWE-bench Lite: a real reported issue in a real open-source project ("PolyFit is not robust to missing data"), with the project's own test as the judge.

```
[repair] fix target inferred from the failure: seaborn/_stats/regression.py
[repair] using the repo's interpreter: .venv
[repair] oracle scope: 1 failing test(s), not the full suite
    [vs] r1 s1/6 t=0.0  gen 17s  verify 10s  ->  PASS

blame {'CODE': 1}   proven slop 0   FIXED True          # 27 seconds total
```

The fix it produced:

```
- p = np.polyfit(x, y, self.order)
+ p = np.polyfit(x.dropna(), y.dropna(), self.order)
```

Three things are worth separating here:

- **The blame is earned.** `CODE`, not `ENVIRONMENT`, not `TEST` — and `proven_slop: 0`, i.e. it did not take the easy exit of declaring the failing test wrong.
- **The oracle is the project's own test**, run with the project's own interpreter. Nothing Determinex wrote decides whether this passed.

- **The control matters.** SWE-bench's official gold patch also passes this oracle ( `1 passed in 0.12s` ), so the target is achievable and a correct fix is recognised as correct. Without that control, a pass proves much less.

Generation ran on the Radeon-hosted 32B; verification ran locally against the repo's tests. `generate()` and `verify()` are decoupled by design, which is what lets the model live on a GPU in another country while the oracle runs where the toolchain is.

**3.4 Self-improving flywheel.** Verified solves are distilled into generalized symptom→fix classes that the fixer applies first on the next occurrence. Seven classes were added to that corpus during this hackathon, each with its method, measured numbers and the traps that produce false readings; all are queryable via `python scripts/determinex_corpus_api.py brief <files>`, which surfaces the ones relevant to whatever a developer is about to edit.

**3.5 Benchmarks — stated honestly.**

| Benchmark | Result | Status |
|---|---|---|
| SWE-bench Lite | 14.0% (42/300) | Audited May-2026 snapshot; clean re-run pending |
| Micro-eval, Engineer v11-dsl | 81.4% (57/70) | Compiler-validated probes |
| Micro-eval, Observer v6-dsl | 75.7% (53/70) | Compiler-validated probes |

Every figure here is compiler-validated or audited. Where a number has not survived a clean re-run it says so, rather than being quoted as settled.

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## 4. Models & Local Deployment

### 4.1 Model family

| Agent | Params | Base | Role |
|---|---|---|---|
| C1 Engineer v11-dsl | 1.5B | Qwen2.5-Coder | Builder |
| C3 Observer v6-dsl | 3B | Llama-3.2 | Monitor |
| C7 Sentinel v5-dsl | 7B | Mistral | Architect |

Published at `huggingface.co/darthceltic85` ( `determinex-engineer` , `determinex-observer-llama-3.2` , `determinex-sentinel` ).

The small sizes are the point: a 1.5B builder is viable *because* the oracle catches its mistakes and verified search retries. Model quality sets `p` ; the system sets `K` .

### 4.2 Deployment on AMD Radeon

Two supported paths. Both reach the same `generate(prompt, temperature) -> str` contract, so nothing downstream knows which produced a candidate.

**Path A — hosted vLLM on a Radeon GPU** (the path measured in this submission):

```
# Radeon Cloud → Add Template → Deploy Type = "vLLM Model API"
vllm serve Qwen/Qwen2.5-Coder-7B-Instruct --host 0.0.0.0 --port 8000

# .env
AMD_BASE_URL=https://.../spaces/<id>/8000/v1
AMD_API_KEY=sk-...

python scripts/determinex_build_from_idea.py --idea idea.md --provider vllm --k 6
```

**Path B — local Ollama on ROCm:**

```
curl -fsSL https://ollama.com/install.sh | sh    # auto-detects ROCm
rocminfo | grep gfx                               # gfx1100 = RDNA3
ollama pull qwen2.5-coder:14b-instruct-q4_K_M
```

**Verification status, stated exactly:** Path A is verified — every number in this document came from it. Path B's ROCm auto-detection is Ollama's documented behaviour and Determinex's accelerator probe covers AMD, but this submission did not run Path B on a physical Radeon card, so it is presented as supported-and-documented, not as measured.

### 4.3 A note on key hygiene

AMD exposes two unrelated endpoints: your own instance, and a shared free endpoint on a third-party host. They now use **two distinct environment variables** ( `AMD_API_KEY` / `AMD_FREE_API_KEY` ). They shared one until this work, which would have transmitted a dedicated-instance key to the shared host on any call routed to the free models. Found while wiring the Radeon endpoint.

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## 5. Optimization for Inference on AMD Radeon GPU

### 5.1 Measured throughput

Qwen2.5-Coder-7B-Instruct, vLLM 0.16.1, ROCm 7.2.1, 512-token completions:

| Mode | Completion tokens | Wall clock | Throughput |
|---|---|---|---|
| Single stream | 512 | 17.05 s | 30.0 tok/s |
| **K=6 concurrent** | 3,072 | **18.41 s** | **166.9 tok/s** |

**5.56× aggregate throughput at 1.08× wall clock.** Measured end-to-end including network.

#### Reproduced three times, on two separate instances

A single measurement of a batching effect is weak evidence, so it was repeated — including on a different instance type, and once on camera:

| Run | Instance | Single | K=6 aggregate | Speedup | Wall-clock cost |
|---|---|---|---|---|---|
| 1 | vLLM Model API endpoint | 30.0 tok/s | 166.9 tok/s | 5.56× | 1.08× |
| 2 | Notebook instance, local vLLM | 31.9 tok/s | 179.0 tok/s | 5.61× | 1.07× |
| 3 | Notebook instance (the demo video) | 32.6 tok/s | 186.9 tok/s | **5.73×** | **1.05×** |

The absolute rates vary with instance load; the *ratio* does not. Across all three, sampling six verified-search candidates cost between 1.05× and 1.08× the wall clock of sampling one, for 5.56–5.73× the aggregate tokens. That stability is the claim — not any single figure.

**Run 6 is the one visible in the demo video** — recorded 2026-08-04 inside the IDE's own terminal against the live instance, so the number on screen and the number in this document can be checked against each other:

| Run | Instance | Single | K=6 aggregate | Speedup | Wall-clock cost |
|---|---|---|---|---|---|
| 6 | Radeon Cloud `u-14786-cdc26df7`, in-IDE terminal | 29.8 tok/s | 166.3 tok/s | 5.58× | 1.07× |

Six runs, five instances, one ratio. The same recording also measures the prefix-cache effect live — **93.5 % hits on the shared-prefix arm against ~1 % when a single distinct token is prepended** — which is §5.2's claim reproduced on camera rather than replayed.

### 5.1a Run 4 — a fourth instance, live on camera, 2026-08-03

Runs 1–3 were captured on a Radeon Cloud instance that was unavailable at the time, so the optimization risked being evidenced only by replay. It was therefore re-measured on **AMD's Radeon Token Factory** while recording: `LIVE_ON_RADEON_2026-08-03.mp4`, 0:55.

| | Value |
|---|---|
| Endpoint | `https://radeon.anruicloud.com/api/v1`, 4 models served |
| Model | MiniCPM5-1B (small, so the sweep finishes inside a recording) |
| Single stream | 64 tokens in 2.45 s = **26.1 tok/s** |
| K sweep | K = 1, 2, 4, 8, 16, 32 — live |
| Selected | **K = 8**, 1.98 candidates/s |

This is a **different ceiling shape from runs 1–3**, and that is the interesting part. The vLLM instance was limited by KV cache and collapsed at K=64. The Token Factory is limited by a published **requests-per-minute quota**: requests above K≈8 are refused outright rather than queued, so the sweep does not find a throughput knee — it finds a wall. Same measurement, same code, three different reasons a machine stops, none of them readable from a config file. That is the argument for measuring in §5.2 made against a third rig.

> **It also found a defect in the calibrator, live.** With the refused points excluded as unusable, the ceiling detector saw only rising throughput among the survivors and printed *"no ceiling within K≤48; this rig has more headroom than the sweep used"* — while 46 of 48 requests had just been refused. Reporting *more* headroom on a rig that had demonstrated less is precisely the class of overclaim this project's oracle discipline exists to prevent, and it was in the tool that measures the GPU. Fixed: when higher-K points were excluded because the backend refused them, that is reported as a ceiling of a different kind, and the corrected wording is what appears in the recording.

### Run 5 — the whole argument, live end to end, 2026-08-03

A fifth instance, and the first where the numbers were measured *while the recording ran* rather than replayed: `LIVE_ON_RADEON_FULL_2min.mp4`, 1:59.

| | |
|---|---|
| Single stream | 256 tokens in 8.67 s = **29.5 tok/s** |
| K=1 | 29.0 agg tok/s |
| K=6 | **163.0 agg tok/s** |
| | **5.61× throughput at 1.07× wall clock** |

That is the fifth independent confirmation of the ratio this whole section rests on — **5.56× · 5.61× · 5.73× · 5.77× · 5.61×**, on five instances, over three weeks. The absolute rates move with instance load; the ratio does not, and the ratio is the claim.

The clip ends on the behaviour that matters more than any throughput figure. Asked to build *"improve the code quality of my project"* — an idea with no input, no output and nothing a test could assert — the system returns `solved=False`: *"no example or typeable invariant could be derived from this idea… that cannot verify behaviour. Add one concrete input/output example and re-run."* Given an idea that can be pinned down, it synthesizes a 3-check oracle and returns a program that passes it, generated on the card. A refusal and a verified solve, from the same model, in ninety seconds.

> **The first take of this clip was re-recorded, and the reason is the point.** It used *"a function called solution that returns the average of a list of numbers"*, which also refused — but for the wrong reason. The spec parser was reading the word after "function" as the function's name, so *"a function called solution"* named it `called`, no example matched, and the oracle came out vacuous. That refusal was a bug wearing the costume of a principle. Fixing the parser (`tests/test_spec_name_extraction.py`) made that same idea solve, which invalidated the take. Demonstrating the real behaviour needs an idea that genuinely cannot be verified — so the clip now uses one. A demo that cannot be reproduced from the shipped code is the exact failure this system exists to prevent.
>
> **What it deliberately does not show,** stated on screen rather than omitted: no `rocm-smi`. This run drives the vLLM endpoint through the instance's own reverse tunnel, not a shell on the box, so the GPU is evidenced by what it serves and how it behaves under load. The hardware readout is in Run 3, which had shell access.

Raw timed terminal log: `amd_gpu_evidence/radeon_live_full_2026-08-03.json`

#### 5.1b The other half of the claim, and where it did not hold

Everything above measures THROUGHPUT: the Kth candidate is nearly free. The argument only completes if spending those candidates converts into correctness — `P = 1 - (1-p)^K` — and that half had never been measured on AMD silicon. It was, on 2026-08-03, against the same live endpoint. The result is not a clean confirmation, which is why it is here in full.

**Finding the regime.** Seven tasks measured p = 1.000 — four textbook (RLE, second-largest, Roman numerals, bracket matching) and three compositions invented for this measurement. A 1B model served on the Radeon one-shots all of them, with distinct completions confirmed, so on that class of work K is pure waste and the calibrator's `p=1 → K=1` rule is exactly right. A saturated task cannot demonstrate amplification, so an eighth was added that needs real reasoning — an arithmetic expression evaluator with precedence, unary minus and truncate-toward-zero division, 6 sound checks:

| | Value |
|---|---|
| Draws | 16 independent, temperature 0.6, **15 of 16 distinct** |
| Passes | 1 |
| **p** | **0.0625** — the productive middle |

**The test, and the miss.**

| | Value |
|---|---|
| K | 8 (the calibrated ceiling for this endpoint, §5.1a) |
| Predicted, i.i.d. | `1 - (1-0.0625)^8` = **0.403** |
| Observed | **1 / 6 trials = 0.167** |

The equation over-predicted. Two candidate explanations, neither dismissable on six trials:

1. **Six trials is not many.** Under a true P = 0.403, seeing ≤1 success in 6 has probability ≈ 0.23. This is consistent with bad luck and is nowhere near evidence against the model.
2. **The draws are not i.i.d., by design.** `p = 0.0625` was measured at a *fixed* temperature 0.6. `VerifiedSearch` deliberately draws its K samples across a temperature LADDER (0.0, 0.2, … 1.0) for diversity, and the first rung is greedy. So a batch of 8 is not eight

draws at p = 0.0625; several sit at temperatures with their own, lower p. The equation assumes a single p and the implementation deliberately violates that to buy coverage.

So the honest statement is: `1 − (1−p)^K` **is an upper estimate for this implementation, not a prediction of it** — and the gap is the price of the diversity mechanism, not evidence the amplifier does not work. It plainly does work: trial 1 had four candidates rejected by the oracle and the fifth pass, which is the machinery visible in one run.

#### 5.1c The gap, closed — measured on the GPU, 2026-08-03

The paragraph above said closing this needed `p` measured *per temperature rung*. The Radeon instance came back, so it was. Qwen2.5-Coder-7B on the live card, 10 draws at every rung of the ladder `VerifiedSearch` actually uses, same 6-check task:

| rung | p | distinct completions |
|---|---|---|
| t=0.0 | **0.00** | **1 / 10** |
| t=0.2 | 0.00 | 2 / 10 |
| t=0.4 | 0.00 | 6 / 10 |
| t=0.6 | 0.00 | 8 / 10 |
| t=0.8 | 0.10 | 6 / 10 |
| t=1.0 | 0.10 | 10 / 10 |

**The per-rung correction was not the answer.** `1 − Π(1−p_i)` = 0.344 against the naive `1−(1−p̄)^8` = 0.337 — the same number. The ladder does not explain the miss.

**What does:** every draw at `t ≤ 0.6` has **p = 0.00**. Four of the eight samples in a K=8 batch cannot pass this task at all, so a **nominal K of 8 is an effective K of 4**. With `p` on the productive rungs anywhere in 0.05–0.10 — and 1/10 is a coarse estimate — the prediction is **0.185–0.344** against **0.167** observed. Consistent.

So `P = 1 − (1−p)^K` was never wrong. **It was being handed the wrong K.** The diversity ladder buys coverage on an easy task and spends half the batch on nothing when the task is hard.

That is now a rule rather than an observation: after a round in which nothing scores, `VerifiedSearch` draws the next round from the hot rungs only ( `_temp_for(..., hot=True)` ). Cold-first remains correct for round one — on an easy task the greedy sample solves it outright and sampling hot would be pure waste.

> **A separate result worth its own line.** `t=0.0` produced **1 distinct completion in 10 draws**. That is the greedy degeneracy which, earlier the same day, made a measurement harness report `112/112, p=1.000` by calling `build_from_idea(k=1)` sixteen times — sixteen draws at temperature zero are one draw. Diagnosed from the source in the morning and independently confirmed on AMD hardware in the evening.

Raw: `amd_gpu_evidence/radeon_p_per_rung_2026-08-03.json`

Raw: `amd_gpu_evidence/radeon_amplification_2026-08-03.json` — all three runs, per-draw, including the saturated set.

> **A harness bug worth recording, since it nearly became the headline.** The first version of this measurement called `build_from_idea(..., k=1)` sixteen times and reported **112/112, p = 1.000** across seven tasks. That was not a result. `VerifiedSearch` 's temperature ladder starts at 0.0, so `k=1` is a GREEDY draw and sixteen of them are one draw sixteen times. A confident number produced by a harness that never varied its input is precisely what the oracle discipline exists to catch, and it was caught only because `distinct_completions` was recorded alongside p. That counter is now part of the artifact.

Raw timed terminal log and the resulting calibration profile: `amd_gpu_evidence/radeon_live_2026-08-03.json` .

### 5.2 Why that number is the whole thesis

Verified search costs K generations per solve. If K generations cost K× the time, the amplifier is a straight latency-for-accuracy trade. On this GPU they cost 1.08×, because vLLM's continuous batching fills the device with concurrent sequences that would otherwise idle during memory-bound decode.

So on Radeon hardware:

| | one sample | six samples |
|---|---|---|
| Wall clock | 17.05 s | 18.41 s (+8%) |
| P(correct) at p=0.3 | 30.0% | **88.2%** |

Correctness nearly triples for an 8% time premium. **This is a GPU property being converted into a correctness property** — it is the concrete answer to "how does this project use the AMD GPU", and it is why the architecture suits this hardware rather than merely running on it.

### 5.2a Choosing K: the scaling curve, and exactly where it breaks

K is the one knob that converts GPU capacity into correctness, so it was swept rather than guessed. Same model, same GPU, 256-token completions:

| K | aggregate tok/s | wall clock | vs K=1 |
|---|---|---|---|
| 1 | 28.7 | 8.92 s | — |
| 8 | 220.3 | 9.30 s | 7.7× tokens, 1.04× time |
| 16 | 413.5 | 9.91 s | 14.4× tokens, 1.11× time |
| **32** | **739.9** | **10.75 s** | **25.8× tokens, 1.21× time** |
| 48 | 849.7 | 13.62 s | 29.6× tokens, 1.53× time |
| 64 | 512.6 ↓ | 29.59 s ↓ | **collapse** |

Throughput scales near-linearly to K=32 while wall clock barely moves. Then K=64 falls off a cliff — throughput drops below K=24 and wall clock nearly triples.

**The cliff is predictable, not mysterious.** vLLM announces its own limit at boot:

```
GPU KV cache size: 506,544 tokens
Maximum concurrency for 8,192 tokens per request: 61.83x
```

K=64 exceeds 61.83, so requests preempt instead of batching. The measured collapse sits exactly where the KV arithmetic says it must. That yields a deployment rule any Radeon owner can apply without re-running this sweep:

> **Keep K below** `floor(kv_cache_tokens / max_model_len)` — read both from vLLM's boot log. Here that is `506544 / 8192 = 61`, and 64 broke.

Correctness follows directly, since `P = 1 - (1 - p)^K`. For a generator right 30% of the time: K=6 → 88.2%, K=16 → 99.7%, **K=32 → 99.999%, for 1.21× the wall clock of a single sample.**

### 5.2b The rule is now code: Determinex calibrates itself to the machine it is on

A deployment rule written in a document is a rule nobody applies. `scripts/determinex_calibrate.py` makes the system derive its own K from the hardware it is actually running on, and `hive/amplifier_bridge.env_k()` uses the result.

It **measures** rather than reads a config value, and that distinction was learned the hard way. vLLM's declared capacity is available programmatically — `vllm:cache_config_info{num_gpu_blocks="31504", block_size="16"}` gives `31504 × 16 ÷ 8192 = 61`, which predicts the collapse at 64 exactly. But the same read-the-config approach applied to Ollama predicted full serialization from `OLLAMA_NUM_PARALLEL=1`, and the machine delivered a 3.4× speedup instead. A client-side config value is not the authority that enforces the limit. So the declared number is used only to **bound the sweep**; the measurement decides.

The objective is **candidates per second** ( `K / wall_clock` ), not aggregate tok/s — because `P = 1 − (1 − p)^K` means what matters is how fast independent attempts arrive, and that metric finds the cliff for free: past the real limit, requests preempt, wall clock explodes, the ratio falls. Run against the captured Radeon sweep it independently selects **K=48** and flags the collapse at **K=64** — reproducing, from the data alone, the knee found by hand.

Two machines, same software, calibrated live:

| Rig | Backend | Optimal K | Candidates/s | Collapse |
|---|---|---|---|---|
| **AMD Radeon Cloud**, ROCm 7.2.1, 7B | vLLM | **48** | 3.52 | K=64 |
| Consumer box, 12 CPU, 1.5B | Ollama | **12** | 1.63 | none within K≤12 |

The old hardcoded `K=6` is wrong on both — it wastes roughly **5×** of the Radeon's correctness rate and about **37%** of the consumer box's. This is the concrete argument for the GPU: the Radeon does not merely run the same workload faster, it sustains **4× the concurrent verification depth**, and depth is what `P` is exponential in.

Uncalibrated rigs get a conservative `K=4` **and a logged line saying so**, naming the command that fixes it — the same doctrine `determinex_oracle` applies to a missing oracle: an absent measurement is reported as absent, never silently guessed. An explicit `DETERMINEX_AMPLIFY_K` always wins; calibration is advice, not a cage.

```
python scripts/determinex_calibrate.py --backend vllm --model Qwen/Qwen2.5-Coder-7B-Instruct
python scripts/determinex_calibrate.py --show
```

Guarded by `tests/test_calibration_k.py` (13 tests), including the negative control that a still-climbing curve is **not** reported as a collapse — otherwise every rig would be capped at whatever K the sweep happened to stop at.

### 5.2c The agent's access pattern is itself a GPU optimization

Verified search issues K requests whose prompts are **identical**, differing only in sampling temperature. That is close to the ideal case for a prefix cache, and unusual — an ordinary multi-request agent sends K *different* prompts. Both arms were measured:

| K | arm | prefix-cache hit rate | aggregate tok/s | wall clock |
|---|---|---|---|---|
| 16 | verified search (shared prefix) | **99.51%** | 410.9 | 9.97 s |
| 16 | distinct prompts | **1.35%** | 292.9 | 13.98 s |
| 32 | verified search (shared prefix) | **99.51%** | 748.5 | 10.73 s |
| 32 | distinct prompts | 49.59% | 517.6 | 15.83 s |

**1.40× the throughput at 0.71× the wall clock, for free**, purely because of how the architecture happens to query the model. `enable_prefix_caching=True` is therefore not a generic nicety here — it is load-bearing for this workload specifically.

> **Method note, because it changed the answer.** The first version of this A/B appended the variant marker to the *end* of each distinct prompt. The long prefix stayed shared, the "distinct" arm scored a 99% hit rate too, and the gap looked like ~30%. Diverging at token 0 is what actually defeats a prefix cache. The corrected arm drops to 1.35%. A control that does not control is worse than no control, so the flawed version is reported here rather than quietly replaced.

Raw data for all three sweeps: `amd_gpu_evidence/k_sweep_and_prefix_cache.json`.

### 5.2d Measuring `p` directly — including where the architecture stops working

`P = 1 − (1 − p)^K` is the load-bearing claim, and `p` is normally assumed. We measured it: **960 independent generations on the Radeon GPU**, 192 per task across five difficulties, every candidate verified against the same synthesized oracle. Sampling at fixed temperature 0.6 so the draws are i.i.d. and `p` is well defined, at concurrency 32 — using this project's own K-sweep result to run its own experiment faster.

| Task | checks | passes / 192 | p |
|---|---|---|---|
| `rle` | 4 | 192/192 | **1.000** |
| `rle` + double-digit run lengths | 6 | 192/192 | **1.000** |
| `gron` (JSON flattening) | 4 | 0/192 | 0.000 |
| `gron` | 5 | 1/192 | 0.005 |
| `gron` + JSON-literal semantics | 6 | 0/192 | **0.000** |

Three things follow, and two of them are uncomfortable.

**1. Our own demo task is trivial.** `rle` — the task in our demo video — is solved by this model **192 times out of 192**, and stays 192/192 after adding double-digit run lengths. Verified search contributes nothing there, because there is nothing to rescue. We would rather state that than let a judge infer it.

**2. Amplification has a floor, and we hit it.** At `p = 0`, `1 − (1 − 0)^K = 0` for every K. No amount of sampling rescues a model that cannot do the task at all. The white-paper phrasing "any `p > 0` is driven toward correct" is exactly right and exactly load-bearing: **verified search multiplies capability, it cannot manufacture it.** That boundary is measured here, not assumed.

**3. Competence looked near-bimodal.** We could not find a task in the middle within budget. Removing the array case from `gron` did not raise `p`, so the barrier is container-declaration semantics rather than indexing. For this model on this family, the transition from 1.000 to ~0.000 is sharp.

> **Honest limit on the curve fit.** At `p = 0.0052` the bootstrap solve-rate curve tracked `1 − (1 − p)^K` to a maximum absolute error of **0.0021** across K=1…32. With a single success in the pool that is close to tautological — resampling one positive nearly reproduces the binomial by construction. It is *consistent with* the law, not strong evidence *for* it. A proper validation needs a task with mid-range `p`, which we did not find in the credits available.

**What this means in practice.** Verified search is worth nothing at `p = 0`, nothing at `p = 1`, and everything in between — but §5.2a shows the *cost* is near-zero out to K=32 (1.21× wall clock for 25.8× the tokens). So the correct deployment policy is not to tune K per task. It is to **run large K always**: nearly free on this hardware, and it pays exactly when `p` lands in the middle, which you cannot know in advance.

Raw data: `amd_gpu_evidence/measured_p_and_amplification.json`.

### 5.2e Two models, one GPU — the three regimes

`p` was then measured for a **second** model on the same hardware, same oracles, same method: 192 independent draws per task, 1,152 generations across both models.

| Task | 7B `p` | 32B `p` | Regime |
|---|---|---|---|
| `rle` | 1.000 | 1.000 | **saturated** — K adds nothing |
| `gron` 5-check | 0.005 | 0.943 | model solves it; K only trims the tail |
| **`gron` 6-check** | **0.000** | **0.490** | **the productive middle** |

*(7B = Qwen2.5-Coder-7B-Instruct bf16; 32B = Qwen2.5-Coder-32B-Instruct-AWQ, 4-bit, ROCm Triton kernels — 18.24 GiB of weights, leaving 23.01 GiB of KV cache.)*

The bottom row is the architecture's whole argument, measured. At `p = 0.490` the model fails **more than half** its attempts:

| K | solve rate | wall-clock cost |
|---|---|---|
| 1 | 48.96% | 1.00× |
| 4 | 93.21% | 1.01× |
| **8** | **99.54%** | **1.04×** |
| 16 | ~100% | 1.11× |

**A coin flip becomes near-certain for 4% more time.** And the top and bottom rows bound the claim honestly: at `p = 1` there is nothing to amplify, and at `p = 0` — the 7B on that same task, 0/192 — no K helps at all, because `1 − (1 − 0)^K = 0`.

> **A rule this produced, applied prospectively.** Serving the 32B at its default 32,768 context reported `Maximum concurrency: 2.88x`, so K=16 would have thrashed exactly like the K=64 collapse in §5.2a. Re-serving at `--max-model-len 4096` gave `94,784 / 4,096 = 23×`, and K=16 ran clean. The bound derived from the K-sweep decided a configuration before it caused a failure, rather than explaining one afterwards.

Raw data: `amd_gpu_evidence/two_model_comparison.json`.

### 5.3 Memory configuration

Telemetry from the running instance:

```
block_size=16 · num_gpu_blocks=31504 · gpu_memory_utilization=0.9 · enable_prefix_caching=True
```

504,064 tokens of KV cache = 26.9 GiB on device (56 KiB/token at 28 layers × 4 KV heads × 128 head-dim, bf16), plus ~14.2 GiB of weights — implying ~46 GiB of Radeon device memory. The headroom is what permits K=6 concurrency without KV eviction; `kv_cache_usage_perc` stayed at 0.0 throughout, so the concurrency measurement was never cache-bound.

`enable_prefix_caching=True` matters more here than in ordinary serving: verified search sends K near-identical prompts, so the shared prefix is computed once and reused K−1 times.

### 5.4 Quantization guidance (Path B)

| GPU | VRAM | Model |
|---|---|---|
| RX 7900 XTX | 24 GB | qwen2.5-coder:32b-instruct-q4_K_M |
| RX 7800 XT | 16 GB | qwen2.5-coder:14b-instruct-q4_K_M |
| RX 7600 | 8 GB | qwen2.5-coder:7b-instruct-q4_K_M |

VRAM fit is arithmetic from published quantization sizes; throughput on these specific cards was not measured and no figures are claimed for them.

### 5.5 Optional bonus criterion: core inference on the Radeon cloud model API, with optimization

The Track 2 rules offer bonus credit for *"core inference running using Radeon cloud model API with quantization or distillation or other optimization methods."* Determinex satisfies this, and each half is separable so a judge can check them independently.

**Core inference on the Radeon cloud model API.** Not a side experiment — the same code path the product ships. `litellm_config.yaml` carries a `radeon/coder7b` role whose endpoint and key are read from the environment (never committed), and any of the four hive roles can be pinned to it for a run:

```
DETERMINEX_VLLM_BASE_URL=https://<instance>/v1  DETERMINEX_VLLM_API_KEY=... \
DETERMINEX_ROLE_BUILDER=radeon/coder7b \
python scripts/determinex_hive.py run-session --session <id>
```

A full session was executed this way, with the builder on the Radeon-hosted model and the Compiler Oracle running in Docker: `[Oracle] Compiler execution backend: docker`, `[AMPLIFY] verified search: PASS`, `Compiler Oracle: PASS`, session complete. Generation on AMD hardware; verification against a real toolchain. Evidence: `amd_gpu_evidence/`.

**The optimization method: measured concurrency calibration.** The optimisation is not a quantization setting copied from a table — it is a *measurement* the system performs on the GPU it is actually given, and the reason it exists is that every fixed answer we tried was wrong on real hardware:

- vLLM's own declared capacity ( `vllm:cache_config_info` ) predicted 15 concurrent requests. The GPU was still scaling linearly at **K=640**. The declaration assumes every request consumes the model's full context; these consumed ~72 tokens. Off by a factor of ~40, so a sweep bounded by that declaration measures a ceiling that does not exist.
- Throughput does not always *collapse* at the ceiling — it can simply **flatten**. Measured: 33.26 candidates/s at K=320 versus 34.17 at K=640. Twice the concurrency, 2.7% more work, double the time-to-first-candidate. A detector that only looks for a fall reports "more headroom" and recommends the worst usable point on the curve.
- The right operating point is therefore derived from the measured curve, not assumed: `python scripts/determinex_calibrate.py --backend vllm --model <m> --base-url <url>` sweeps K, detects collapse *and* saturation, and stores the curve per machine. An uncalibrated rig is told it is uncalibrated rather than handed a confident default.

Quantization is used where it is the right tool — the 32B is served AWQ-quantized to fit alongside a second model (§5.2e, §5.4) — but the optimisation this project contributes is the calibration itself, because it transfers: it makes the *next* Radeon GPU fast without anyone re-deriving the constant by hand.

**What is deliberately not claimed.** Cloud models from other vendors are supported by the provider registry and were used during development, but they are not core: the shipped default is a local Ollama model, and the Radeon path is the AMD-hosted alternative. The rules state that core inference must run on AMD hardware, and nothing in the verified path depends on a non-AMD API.

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## 6. Boundaries — what this does not do

§2 states the rule this project is built on: *a check must never report an outcome it did not establish.* Applied to a submission, that means the limits belong in it, at the same volume as the results. Every item below is measured or verified, not hedging.

**Verified search cannot manufacture capability.** At `p = 0`, `1 − (1−0)^K = 0` for every K. Measured, not assumed: the 7B scored **0/192** on `gron` with JSON-literal semantics, and no amount of sampling rescues it (§5.2d). The architecture multiplies a model's competence; it does not create it. At `p = 1` it is equally worthless — our own demo task, `rle`, is **192/192** one-shot, so verified search contributes nothing there. It pays only in the middle.

**A wrong oracle produces confident wrong answers.** Soundness is load-bearing and not automatic. This is why a synthesized oracle that can only assert "the symbol exists" is rejected as vacuous, why every oracle refuses an empty workspace, and why the Test Validator declares a test slop only with proof. But an oracle that is sound *and wrong about the intent* is not detectable from inside the system, and we do not claim otherwise.

**The SWE-bench figure is a snapshot, not a settled result.** 14.0% comes from an audited May-2026 run whose clean re-run has not completed, so it is quoted as provisional. §3.3a's repair is a single instance solved end to end with a control — it demonstrates the loop, it is not a benchmark score.

**Path B is documented, not measured.** ROCm auto-detection via Ollama is Ollama's documented behaviour and our accelerator probe covers AMD, but no physical Radeon card ran Path B for this submission. The VRAM table in §5.4 is arithmetic from published quantization sizes — no throughput is claimed for those cards.

**One live run shows no `rocm-smi`.** Run 5 drives the vLLM endpoint through the instance's own reverse tunnel rather than a shell on the box. The driver readout is in Run 3, which had shell access; Run 5's hardware evidence is what it serves and how it behaves under load.

**The p/K validation is consistent, not conclusive.** §5.1c resolves the over-prediction — nominal K ≠ effective K — and the corrected estimate brackets the observation. But `p` per rung came from 10 draws, and 6 trials is a small denominator. It is consistent with the law, not strong evidence for it.

**Cloak's leak count of 0 means one specific thing.** It proves everything Cloak obfuscated stayed obfuscated. Extraction *completeness* is a separate property with its own test, and that test has caught real gaps — JavaScript instance fields written `this.field = 0` survived obfuscation until fixed. Cloaked runs over JS repositories predating that fix are not covered by the audit's guarantee.

**Three execution sites remain flagged for owner decision.** The execution-layer audit reports 515 sites with 0 unsafe and 0 requiring migration, but 3 sites in the agent registry — a coding agent CLI spawn and two auth probes — are trusted-by-design rather than sandboxed. They are left flagged rather than silently exempted.

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